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THE UNIVERSITY OF HONG KONG
DEPARTMENT OF CHEMISTRY

CHEM1043 General Chemistry II

May 18, 2021 (Tuesday) **2:30 pm – 4:30 pm**

(Cutoff time for uploading your answer script to OLEX: **4:45 pm**)

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

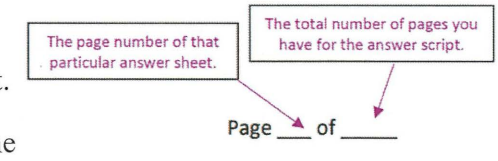
Candidates are permitted to refer to the following electronic/printed/handwritten materials in the examination: textbook, lecture slides, assignment handout, and self-made notes. Internet searching and crowdsourcing from group messages, online forums or social media, etc. are strictly forbidden.

This paper consists of **six** questions (100 marks). Answer **ALL** questions.
Hand-write your answers on A4 size white paper using black pen or blue pen.
Write on each answer sheet your student number and the page number.

A list of physical constants and a periodic table are provided on the last pages of this question paper.

At the end of the examination, please do the following:

- ☐ Check that you have written your **student number** and **page number** on each page of your answer script.
- ☐ Scan or take photos of your answer script and combine the pages into one single file.
- ☐ Check your file to make sure all pages are included and the images are clear.
- ☐ Submit your file through OLEX. And **make sure you have submitted the correct and complete file**.



There are **six questions** in this paper. Answer **ALL** questions. **Hand-write** your answers on A4 paper.

Question 1 (17 marks)

(a) A mixture of gases, weighed 0.428 g, is contained in a vessel at 1.75 atm. The mixture is found to consist of 15.6 % $N_2(g)$, 46.0 % N_2O (g), and 38.4 % CO_2 (g) by mass. What is the partial pressure of each gas in the mixture? Show your calculations.

(6 marks)

(b) Consider the following reaction.

$$2KClO_3\ (s) \rightarrow 2KCl\ (s) + 3O_2\ (g)$$

A 0.250 L flask is filled with oxygen that has been collected over water at 14 °C and an atmospheric pressure of 729 Torr (see Figure 1 below).

Calculate the molar volume of O_2 (g) at 0 °C and 760 Torr. The vapor pressure of water at 14 °C is 12.0 Torr. Show your calculations.

(5 marks)

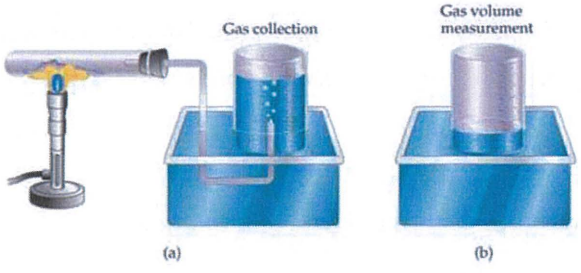


Figure 1

(a) Collection of gas over water.

(b) When the gas has been collected, the bottle is adjusted so that the heights of the water inside and outside the collection vessel are equalized.

(c) The root mean square speed, u_{rms}, of helium gas at 250 K is 1.25×10^3 m/s.

Show your calculations for the following. You may refer to the List of Physical Constants for the equations concerning kinetic theory of gases.

(i) What is the average kinetic energy of an atom in helium gas at 250 K?

(ii) What is u_{rms} of oxygen (O_2) gas at the same temperature?

(iii) If a molecule in nitrogen gas has the same u_{rms} as an atom in helium at 250 K, what is the temperature of the N_2 gas?

(6 marks)

Question 2 (19 marks)

(a) (i) Discuss the difference between a σ-bond and a π-bond.

(3 marks)

(ii) Account for the differences between a bonding molecular orbital and an antibonding molecular orbital as far as you can.

(3 marks)

(iii) Explain how the bond order relates to the bond length and the bond strength, in general.

(3 marks)

(iv) Represent the bonding in the cyanide ion, CN^-, with a labelled molecular orbital diagram showing the atomic orbitals involved and the molecular orbitals formed. Hence, state the bond order and magnetic property (i.e. diamagnetic or paramagnetic). Give your reasoning.

(6 marks)

(b) Use band theory to explain the electric conductivity of semi-conductors.

(4 marks)

Question 3 (18 marks)

(a) The minimum mass concentration of oxygen required for fish life is 4 mg/L. The Henry's law constant for oxygen in water at 20 °C is 1.3×10^{-3} mol L^{-1} atm^{-1}.

(i) Assuming the density of a solution is 1.00 g/mL, express this concentration in parts per million (ppm).

(ii) What is the minimum partial pressure of O_2 that would supply this concentration in water at 20.0 °C?

(iii) What is the minimum atmospheric pressure that would give this partial pressure, assuming oxygen exerts 21 % of the atmospheric pressure?

(6 marks)

(b) The normal boiling point of ethanol is 78.4 °C. When 9.15 g of a soluble non-volatile non-electrolyte was dissolved in 100.0 g of ethanol, the vapour pressure of the solution at that temperature (78.4 °C) was 740 torr. Take 1 atm = 760 torr.

(i) What are the mole fractions of the ethanol and the solute?

(ii) What is the molar mass of the solute?

(6 marks)

(Con't) Question 3

(c) The addition of 0.24 g of sulfur to 100.0 g of CCl_4 lowers the freezing point of CCl_4 by 0.28 °C. The freezing-point constant of CCl_4 is 29.8 °C mol^{-1} kg. Determine the molar mass and molecular formula of sulfur. Show your calculations and reasoning.

(6 marks)

Question 4 (16 marks)

(a) Discuss the distinction between the two terms, *solubility* and *solubility product constant*.

(3 marks)

(b) $MgCO_3$ has $K_{sp} = 1.0 \times 10^{-5}$ and $Ni(OH)_2$ has $K_{sp} = 6.5 \times 10^{-18}$. Calculate the molar solubility of each sparingly soluble substance in its corresponding solution:

(i) magnesium carbonate in 4.2×10^{-5} M Na_2CO_3 (aq);

(ii) nickel hydroxide in 3.7×10^{-5} M $NiSO_4$ (aq).

Show your working.

(7 marks)

(c) Suppose we add 0.150 L of 0.100 M $MgCl_2$ to 0.350 L of 0.150 M NH_3 solution at 25 °C. To prevent precipitation of $Mg(OH)_2$ (s) from this solution, what mass (in g) of $(NH_4)_2SO_4$ should be added?

$Mg(OH)_2$ has $K_{sp} = 1.8 \times 10^{-11}$ and NH_3 has $K_b = 1.8 \times 10^{-5}$. The molar mass of $(NH_4)_2SO_4$ is 132.1 g mol^{-1}. Show your calculations and reasoning.

(6 marks)

To be continued

Question 5 (18 marks)

Consider the following reaction,

$$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$$

The values of K_p are 2.15×10^{11} at 200 °C and 4.56×10^8 at 260 °C, respectively.

(i) Determine $\Delta H°$ by using the van't Hoff equation.

van't Hoff equation: $\ln\frac{K_2}{K_1} = \frac{\Delta H^o}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$ (6 marks)

(ii) Determine $\Delta H°$ by using the tabulated data in **Appendix 1** attached to this exam paper. (4 marks)

(iii) Compare the two results from parts (i) and (ii) above and comment on how good is the assumption that $\Delta H°$ is essentially independent of temperature in this case between 200 °C and 260 °C.

(2 marks)

(iv) Use data in **Appendix 1** attached to this exam paper to determine the values of $\Delta G°$ and K (equilibrium constant) at 298 K for the following reaction (note you may need to balance the reaction equation).

$$Ag^+(aq) + SO_4^{2-}(aq) \rightleftharpoons Ag_2SO_4(s)$$

(6 marks)

To be continued

Question 6 (12 marks)

The Silver-Zinc Cell (also known as a button battery) is commonly used in watches, hearing aids and cameras. The amount of electrolyte can be small and the electrodes can be very close together to give a very high energy density suitable for small batteries. The silver-zinc battery has a cell diagram as follows:

$Zn(s), ZnO(s) \mid KOH\ (satd) \mid Ag_2O(s), Ag(s)$

The half-cell and overall cell reactions are:

Reduction:	$Ag_2O\,(s) + H_2O\,(l) + 2e^-$	$\rightleftharpoons$	$2Ag\,(s) + 2OH^-\,(aq)$
Oxidation:	$Zn\,(s) + 2OH^-\,(aq)$	$\rightleftharpoons$	$ZnO\,(s) + H_2O\,(l) + 2e^-$
Overall Reaction:	$Zn\,(s) + Ag_2O\,(s)$	$\rightleftharpoons$	$ZnO\,(s) + 2Ag\,(s)$

Use the thermodynamic data from **Appendix 1** attached to this exam paper to calculate a theoretical voltage for the silver-zinc battery described in this question (assume the reaction occurs at 298 K).

Hint: First find ΔG° for the overall reaction. (6 marks)

Then find E°_{cell}. (6 marks)

Show your calculations.

– END OF PAPER –

Appendix 1

Thermodynamic Properties of Substances at 298.15 K.[a] Substances are at 1 bar pressure. For aqueous solutions, solutes are at unit activity (roughly 1 M). Data for ions in aqueous solution are relative to values of zero for $\Delta_f H°$, $\Delta_f G°$, and $S°$ for H^+.

[a]Data for inorganic substances and for organic compounds with up to two carbon atoms per molecule are adapted from D. D. Wagman, et al., "The NBS Tables of Chemical Thermodynamic Properties: Selected Values for Inorganic and C_1 and C_2 Organic Substances in SI Units" *Journal of Physical and Chemical Reference Data* 11 (1982) Supplement 2. Data for other organic compounds are from J. A. Dean, *Lange's Handbook of Chemistry*. 15th ed., McGraw-Hill, 1999, and other sources.

Inorganic Substances				
	$\Delta_f H°$, kJ mol^{-1}	$\Delta_f G°$, kJ mol^{-1}	$S°$, J mol^{-1} K^{-1}	C_p, J mol^{-1} K^{-1}
Carbon (See also the table of organic substances.)				
C(g)	716.7	671.3	158.0	20.8
C(diamond)	1.90	2.90	2.38	6.1
C(graphite)	0	0	5.74	8.5
CCl_4(g)	−102.9	−60.59	309.9	83.3
CCl_4(l)	−135.4	−65.21	216.4	130.7
C_2N_2(g)	309.0	297.4	241.9	56.8
CO(g)	−110.5	−137.2	197.7	29.1
CO_2(g)	−393.5	−394.4	213.7	37.1
CO_3^{2-}(aq)	−677.1	−527.8	−56.9	—
C_3O_2(g)	−93.72	−109.8	276.5	67.0
C_3O_2(l)	−117.3	−105.0	181.1	—
$COCl_2$(g)	−218.8	−204.6	283.5	57.7
COS(g)	−142.1	−169.3	231.6	41.5
CS_2(l)	89.70	65.27	151.3	76.4
Hydrogen				
H(g)	218.0	203.2	114.7	20.8
H^+(aq)	0	0	0	0
H_2(g)	0	0	130.7	28.8
HBr(g)	−36.40	−53.45	198.7	29.1
HCl(g)	−92.31	−95.30	186.9	29.1
HCl(aq)	−167.2	−131.2	56.5	−136.4
$HClO_2$(aq)	−51.9	5.9	188.3	—
HCN(g)	135.1	124.7	201.8	35.9
HF(g)	−271.1	−273.2	173.8	—
HI(g)	26.48	1.70	206.6	29.2
HNO_3(l)	−174.1	−80.71	155.6	109.9
HNO_3(aq)	−207.4	−111.3	146.4	−86.6
H_2O(g)	−241.8	−228.6	188.8	33.6
H_2O(l)	−285.8	−237.1	69.91	75.3
H_2O_2(g)	−136.3	−105.6	232.7	43.1
H_2O_2(l)	−187.8	−120.4	109.6	89.1
H_2S(g)	−20.63	−33.56	205.8	34.2
H_2SO_4(l)	−814.0	−690.0	156.9	138.9
H_2SO_4(aq)	−909.3	−744.5	20.1	−293.0
Silver				
Ag(s)	0	0	42.55	25.4
Ag^+(aq)	105.6	77.11	72.68	21.8
AgBr(s)	−100.4	−96.90	107.1	52.4
AgCl(s)	−127.1	−109.8	96.2	50.8
AgI(s)	−61.84	−66.19	115.5	56.8
$AgNO_3$(s)	−124.4	−33.41	140.9	93.1
Ag_2O(s)	−31.05	−11.20	121.3	65.9
Ag_2SO_4(s)	−715.9	−618.4	200.4	131.4

To be continued

Appendix 1 (Con't)

Inorganic Substances				
	$\Delta_f H°$, kJ mol^{-1}	$\Delta_f G°$, kJ mol^{-1}	$S°$, J mol^{-1} K^{-1}	C_p, J mol^{-1} K^{-1}
Sulfur				
S(g)	278.8	238.3	167.8	23.7
S(rhombic)	0	0	31.80	22.6
S_8(g)	102.3	49.63	431.0	156.06
S_2Cl_2(g)	−18.4	−31.8	331.5	124.3
SF_6(g)	−1209	−1105	291.8	97.0
SO_2(g)	−296.8	−300.2	248.2	39.9
SO_3(g)	−395.7	−371.1	256.8	50.7
SO_4^{2-}(aq)	−909.3	−744.5	20.1	−293.0
$S_2O_3^{2-}$(aq)	−648.5	−522.5	67	—
SO_2Cl_2(g)	−364.0	−320.0	311.9	77.0
SO_2Cl_2(l)	−394.1	—	—	−134.0
Zinc				
Zn(s)	0	0	41.63	25.4
Zn^{2+}(aq)	−153.9	−147.1	112.1	46.0
ZnO(s)	−348.3	−318.3	43.64	40.3

Organic Substances					
	Name	$\Delta_f H°$, kJ mol^{-1}	$\Delta_f G°$, kJ mol^{-1}	$S°$, J mol^{-1} K^{-1}	C_p, J mol^{-1} K^{-1}
CH_4(g)	Methane(g)	−74.81	−50.72	186.3	35.7
C_2H_2(g)	Acetylene(g)	226.7	209.2	200.9	44.0
C_2H_4(g)	Ethylene(g)	52.26	68.15	219.6	42.9
C_2H_6(g)	Ethane(g)	−84.68	−32.82	229.6	52.5
C_3H_8(g)	Propane(g)	−103.8	−23.3	270.3	73.6
C_4H_{10}(g)	Butane(g)	−125.6	−17.1	310.2	97.5
C_6H_6(g)	Benzene(g)	82.6	129.8	269.3	82.4
C_6H_6(l)	Benzene(l)	49.0	124.5	173.4	136.0
C_6H_{12}(g)	Cyclohexane(g)	−123.4	32.0	298.4	106.3
C_6H_{12}(l)	Cyclohexane(l)	−156.4	26.9	204.4	154.9
$C_{10}H_8$(g)	Naphthalene(g)	150.6	224.2	333.2	131.9
$C_{10}H_8$(s)	Naphthalene(s)	77.9	201.7	167.5	165.7
CH_2O(g)	Formaldehyde(g)	−108.6	−102.5	218.8	35.4
CH_3CHO(g)	Acetaldehyde(g)	−166.2	−128.9	250.3	55.3
CH_3CHO(l)	Acetaldehyde(l)	−192.3	−128.1	160.2	89.0
CH_3OH(g)	Methanol(g)	−200.7	−162.0	239.8	44.1
CH_3OH(l)	Methanol(l)	−238.7	−166.3	126.8	81.1
CH_3CH_2OH(g)	Ethanol(g)	−235.1	−168.5	282.7	65.6
CH_3CH_2OH(l)	Ethanol(l)	−277.7	−174.8	160.7	112.3
C_6H_5OH(s)	Phenol(s)	−165.1	−50.4	144.0	127.4
$(CH_3)_2CO$(g)	Acetone(g)	−216.6	−153.0	295.0	74.5
$(CH_3)_2CO$(l)	Acetone(l)	−247.6	−155.6	200.5	126.3
CH_3COOH(g)	Acetic acid(g)	−432.3	−374.0	282.5	63.4
CH_3COOH(l)	Acetic acid(l)	−484.5	−389.9	159.8	123.3
CH_3COOH(aq)	Acetic acid(aq)	−485.8	−396.5	178.7	−6.3
C_6H_5COOH(s)	Benzoic acid(s)	−385.2	−245.3	167.6	146.8
CH_3NH_2(g)	Methylamine(g)	−22.97	32.16	243.4	50.1
$C_6H_5NH_2$(g)	Aniline(g)	86.86	166.8	319.3	107.9
$C_6H_5NH_2$(l)	Aniline(l)	31.6	149.2	191.3	191.9

To be continued

Appendix 1 (Con't)

Standard Electrode (Reduction) Potentials at 25 °C

Reduction half-reaction	$E°$, V
$F_2(g) + 2\,e^- \longrightarrow 2\,F^-(aq)$	+2.866
$OF_2(g) + 2\,H^+(aq) + 4\,e^- \longrightarrow H_2O(l) + 2\,F^-(aq)$	+2.1
$O_3(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow O_2(g) + H_2O(l)$	+2.075
$S_2O_8^{2-}(aq) + 2\,e^- \longrightarrow 2\,SO_4^{2-}(aq)$	+2.01
$Ag^{2+}(aq) + e^- \longrightarrow Ag^+(aq)$	+1.98
$H_2O_2(aq) + 2\,H^+(aq) + 2\,e^- \longrightarrow 2\,H_2O(l)$	+1.763
$MnO_4^-(aq) + 4\,H^+(aq) + 3\,e^- \longrightarrow MnO_2(s) + 2\,H_2O(l)$	+1.70
$PbO_2(s) + SO_4^{2-}(aq) + 4\,H^+(aq) + 2\,e^- \longrightarrow PbSO_4(s) + 2\,H_2O(l)$	+1.69
$Au^{3+}(aq) + 3\,e^- \longrightarrow Au(s)$	+1.52
$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)$	+1.51
$2\,BrO_3^-(aq) + 12\,H^+(aq) + 10\,e^- \longrightarrow Br_2(l) + 6\,H_2O(l)$	+1.478
$PbO_2(s) + 4\,H^+(aq) + 2\,e^- \longrightarrow Pb^{2+}(aq) + 2\,H_2O(l)$	+1.455
$ClO_3^-(aq) + 6\,H^+(aq) + 6\,e^- \longrightarrow Cl^-(aq) + 3\,H_2O(l)$	+1.450
$Au^{3+}(aq) + 2\,e^- \longrightarrow Au^+(aq)$	+1.36
$Cl_2(g) + 2\,e^- \longrightarrow 2\,Cl^-(aq)$	+1.358
$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^- \longrightarrow 2\,Cr^{3+}(aq) + 7\,H_2O(l)$	+1.33
$MnO_2(s) + 4\,H^+(aq) + 2\,e^- \longrightarrow Mn^{2+}(aq) + 2\,H_2O(l)$	+1.23
$O_2(g) + 4\,H^+(aq) + 4\,e^- \longrightarrow 2\,H_2O(l)$	+1.229
$2\,IO_3^-(aq) + 12\,H^+(aq) + 10\,e^- \longrightarrow I_2(s) + 6\,H_2O(l)$	+1.20
$ClO_4^-(aq) + 2\,H^+(aq) + 2\,e^- \longrightarrow ClO_3^-(aq) + H_2O(l)$	+1.189
$ClO_3^-(aq) + 2\,H^+(aq) + e^- \longrightarrow ClO_2(g) + H_2O(l)$	+1.175
$NO_2(g) + H^+(aq) + e^- \longrightarrow HNO_2(aq)$	+1.07
$Br_2(l) + 2\,e^- \longrightarrow 2\,Br^-(aq)$	+1.065
$NO_2(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow NO(g) + H_2O(l)$	+1.03
$[AuCl_4]^-(aq) + 3\,e^- \longrightarrow Au(s) + 4\,Cl^-(aq)$	+1.002
$VO_2^+(aq) + 2\,H^+(aq) + e^- \longrightarrow VO^{2+}(aq) + H_2O(l)$	+1.000
$NO_3^-(aq) + 4\,H^+(aq) + 3\,e^- \longrightarrow NO(g) + 2\,H_2O(l)$	+0.956
$Cu^{2+}(aq) + I^-(aq) + e^- \longrightarrow CuI(s)$	+0.86
$Hg^{2+}(aq) + 2\,e^- \longrightarrow Hg(l)$	+0.854
$Ag^+(aq) + e^- \longrightarrow Ag(s)$	+0.800
$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$	+0.771
$O_2(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow H_2O_2(aq)$	+0.695
$2\,HgCl_2(aq) + 2\,e^- \longrightarrow Hg_2Cl_2(s) + 2\,Cl^-(aq)$	+0.63
$MnO_4^-(aq) + e^- \longrightarrow MnO_4^{2-}(aq)$	+0.56
$I_2(s) + 2\,e^- \longrightarrow 2\,I^-(aq)$	+0.535
$Cu^+(aq) + e^- \longrightarrow Cu(s)$	+0.520
$H_2SO_3(aq) + 4\,H^+(aq) + 4\,e^- \longrightarrow S(s) + 3\,H_2O(l)$	+0.449
$C_2N_2(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow 2\,HCN(aq)$	+0.37
$[Fe(CN)_6]^{3-}(aq) + e^- \longrightarrow [Fe(CN)_6]^{4-}(aq)$	+0.361
$VO^{2+}(aq) + 2\,H^+(aq) + e^- \longrightarrow V^{3+}(aq) + H_2O(l)$	+0.337
$Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$	+0.340
$PbO_2(s) + 2\,H^+(aq) + 2\,e^- \longrightarrow PbO(s) + H_2O(l)$	+0.28
$Hg_2Cl_2(s) + 2\,e^- \longrightarrow 2\,Hg(l) + 2\,Cl^-(aq)$	+0.2676
$HAsO_2(aq) + 3\,H^+(aq) + 3\,e^- \longrightarrow As(s) + 2\,H_2O(l)$	+0.240
$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$	+0.2223
$SO_4^{2-}(aq) + 4\,H^+(aq) + 2\,e^- \longrightarrow 2\,H_2O(l) + SO_2(g)$	+0.17
$Cu^{2+}(aq) + e^- \longrightarrow Cu^+(aq)$	+0.159
$Sn^{4+}(aq) + 2\,e^- \longrightarrow Sn^{2+}(aq)$	+0.154
$S(s) + 2\,H^+(aq) + 2\,e^- \longrightarrow H_2S(g)$	+0.144
$AgBr(s) + e^- \longrightarrow Ag(s) + Br^-(aq)$	+0.071
$2\,H^+(aq) + 2\,e^- \longrightarrow H_2(g)$	0
$Pb^{2+}(aq) + 2\,e^- \longrightarrow Pb(s)$	−0.125
$Sn^{2+}(aq) + 2\,e^- \longrightarrow Sn(s)$	−0.137
$AgI(s) + e^- \longrightarrow Ag(s) + I^-(aq)$	−0.152
$V^{3+}(aq) + e^- \longrightarrow V^{2+}(aq)$	−0.255
$Ni^{2+}(aq) + 2\,e^- \longrightarrow Ni(s)$	−0.257
$H_3PO_4(aq) + 2\,H^+(aq) + 2\,e^- \longrightarrow H_3PO_3(aq) + H_2O(l)$	−0.276

To be continued

Appendix 1 (Con't)

Reduction half-reaction	$E°$, V
$Co^{2+}(aq) + 2\,e^- \longrightarrow Co(s)$	−0.277
$In^{3+}(aq) + 3\,e^- \longrightarrow In(s)$	−0.338
$PbSO_4(s) + 2\,e^- \longrightarrow Pb(s) + SO_4^{2-}(aq)$	−0.356
$Cd^{2+}(aq) + 2\,e^- \longrightarrow Cd(s)$	−0.403
$Cr^{3+}(aq) + e^- \longrightarrow Cr^{2+}(aq)$	−0.424
$Fe^{2+}(aq) + 2\,e^- \longrightarrow Fe(s)$	−0.440
$2\,CO_2(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow H_2C_2O_4(aq)$	−0.49
$Zn^{2+}(aq) + 2\,e^- \longrightarrow Zn(s)$	−0.763
$Cr^{2+}(aq) + 2\,e^- \longrightarrow Cr(s)$	−0.90
$Mn^{2+}(aq) + 2\,e^- \longrightarrow Mn(s)$	−1.18
$Ti^{2+}(aq) + 2\,e^- \longrightarrow Ti(s)$	−1.63
$U^{3+}(aq) + 3\,e^- \longrightarrow U(s)$	−1.66
$Al^{3+}(aq) + 3\,e^- \longrightarrow Al(s)$	−1.676
$Mg^{2+}(aq) + 2\,e^- \longrightarrow Mg(s)$	−2.356
$La^{3+}(aq) + 3\,e^- \longrightarrow La(s)$	−2.38
$Na^+(aq) + e^- \longrightarrow Na(s)$	−2.713
$Ca^{2+}(aq) + 2\,e^- \longrightarrow Ca(s)$	−2.84
$Sr^{2+}(aq) + 2\,e^- \longrightarrow Sr(s)$	−2.89
$Ba^{2+}(aq) + 2\,e^- \longrightarrow Ba(s)$	−2.92
$Cs^+(aq) + e^- \longrightarrow Cs(s)$	−2.923
$K^+(aq) + e^- \longrightarrow K(s)$	−2.924
$Rb^+(aq) + e^- \longrightarrow Rb(s)$	−2.924
$Li^+(aq) + e^- \longrightarrow Li(s)$	−3.040
Basic solution	
$O_3(g) + H_2O(l) + 2\,e^- \longrightarrow O_2(g) + 2\,OH^-(aq)$	+1.246
$ClO^-(aq) + H_2O(l) + 2\,e^- \longrightarrow Cl^-(aq) + 2\,OH^-(aq)$	+0.890
$H_2O_2(aq) + 2\,e^- \longrightarrow 2\,OH^-(aq)$	+0.88
$BrO^-(aq) + H_2O(l) + 2\,e^- \longrightarrow Br^-(aq) + 2\,OH^-(aq)$	+0.766
$ClO_3^-(aq) + 3\,H_2O(l) + 6\,e^- \longrightarrow Cl^-(aq) + 6\,OH^-(aq)$	+0.622
$2\,AgO(s) + H_2O(l) + 2\,e^- \longrightarrow Ag_2O(s) + 2\,OH^-(aq)$	+0.604
$MnO_4^-(aq) + 2\,H_2O(l) + 3\,e^- \longrightarrow MnO_2(s) + 4\,OH^-(aq)$	+0.60
$BrO_3^-(aq) + 3\,H_2O(l) + 6\,e^- \longrightarrow Br^-(aq) + 6\,OH^-(aq)$	+0.584
$2\,BrO^-(aq) + 2\,H_2O(l) + 2\,e^- \longrightarrow Br_2(l) + 4\,OH^-(aq)$	+0.455
$2\,IO^-(aq) + 2\,H_2O(l) + 2\,e^- \longrightarrow I_2(s) + 4\,OH^-(aq)$	+0.42
$O_2(g) + 2\,H_2O(l) + 4\,e^- \longrightarrow 4\,OH^-(aq)$	+0.401
$Ag_2O(s) + H_2O(l) + 2\,e^- \longrightarrow 2\,Ag(s) + 2\,OH^-(aq)$	+0.342
$Co(OH)_3(s) + e^- \longrightarrow Co(OH)_2(s) + OH^-(aq)$	+0.17
$2\,MnO_2(s) + H_2O(l) + 2\,e^- \longrightarrow Mn_2O_3(s) + 2\,OH^-(aq)$	+0.118
$NO_3^-(aq) + H_2O(l) + 2\,e^- \longrightarrow NO_2^-(aq) + 2\,OH^-(aq)$	+0.01
$CrO_4^{2-}(aq) + 4\,H_2O(l) + 3\,e^- \longrightarrow Cr(OH)_3(s) + 5\,OH^-(aq)$	−0.11
$S(s) + 2\,e^- \longrightarrow S^{2-}(aq)$	−0.48
$HPbO_2^-(aq) + H_2O(l) + 2\,e^- \longrightarrow Pb(s) + 3\,OH^-(aq)$	−0.54
$HCHO(aq) + 2\,H_2O(l) + 2\,e^- \longrightarrow CH_3OH(aq) + 2\,OH^-(aq)$	−0.59
$SO_3^{2-}(aq) + 3\,H_2O(l) + 4\,e^- \longrightarrow S(s) + 6\,OH^-(aq)$	−0.66
$AsO_4^{3-}(aq) + 2\,H_2O(l) + 2\,e^- \longrightarrow AsO_2^-(aq) + 4\,OH^-(aq)$	−0.67
$AsO_2^-(aq) + 2\,H_2O(l) + 3\,e^- \longrightarrow As(s) + 4\,OH^-(aq)$	−0.68
$Cd(OH)_2(s) + 2\,e^- \longrightarrow Cd(s) + 2\,OH^-(aq)$	−0.824
$2\,H_2O(l) + 2\,e^- \longrightarrow H_2(g) + 2\,OH^-(aq)$	−0.828
$OCN^-(aq) + H_2O(l) + 2\,e^- \longrightarrow CN^-(aq) + 2\,OH^-(aq)$	−0.97
$As(s) + 3\,H_2O(l) + 3\,e^- \longrightarrow AsH_3(g) + 3\,OH^-(aq)$	−1.21
$Zn(OH)_2(s) + 2\,e^- \longrightarrow Zn(s) + 2\,OH^-(aq)$	−1.246
$Sb(s) + 3\,H_2O(l) + 3\,e^- \longrightarrow SbH_3(g) + 3\,OH^-(aq)$	−1.338
$Al(OH)_4^-(aq) + 3\,e^- \longrightarrow Al(s) + 4\,OH^-(aq)$	−2.310
$Mg(OH)_2(s) + 2\,e^- \longrightarrow Mg(s) + 2\,OH^-(aq)$	−2.687

End of Appendix 1

Physical Constants

Planck's constant	$h = 6.626 \times 10^{-34}$ J s molecule^{-1}
Velocity of light in vacuum	$c = 2.998 \times 10^{8}$ m s^{-1}
Avogadro's number	$N = 6.022 \times 10^{23}$ molecules mol^{-1}
Boltzmann constant	$k = 1.381 \times 10^{-23}$ J K^{-1} molecule^{-1}
Gas constant	$R = 8.314$ J K^{-1} mol^{-1}
	$= 0.08206$ atm dm^3 K^{-1} mol^{-1}
	$= 0.08314$ bar dm^3 K^{-1} mol^{-1}
	$= 8.314$ Pa m^3 K^{-1} mol^{-1}
Faraday constant	$F = 96487$ C mol^{-1} (of electrons transferred)
	$= 23061$ cal volt^{-1} mol^{-1} (of electrons transferred)
Elementary charge	$e = 1.602 \times 10^{-19}$ C
Mass of electron	$m_e = 9.109 \times 10^{-31}$ kg
Mass of proton	$m_p = 1.673 \times 10^{-27}$ kg
Standard gravitational acceleration	$g = 9.807$ m s^{-2}
Ionic product of water	$K_w = 1.0 \times 10^{-14}$ (at 25 °C)

Some Equations for Kinetic Theory of Gases:			
$u_{rms} = \sqrt{\frac{3RT}{M}}$	$\overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$	$\overline{E_k} = \frac{1}{2}m\overline{u^2}$	$P = \frac{1}{3}\frac{N}{V}m\overline{u^2}$

Conversion Table:	**Numerical Constants:**
Specific heat capacity of water = 1 cal K^{-1} g^{-1}	$\pi = 3.141593$
$= 4.184$ J K^{-1} g^{-1}	
1 erg = 10^{-7} J	e = 2.718282
1 G = 10^{-4} T (G = gauss, T = tesla)	ln x = 2.303 log x
1 atm = 101,325 Pa = 760 mmHg (Torr)	
1 bar = 10^5 Pa	
0 °C = 273.15 K	

The periodic table

1	2		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Hydrogen 1 **H** 1.008																		Helium 2 **He** 4.0026
Lithium 3 **Li** 6.94	Beryllium 4 **Be** 9.0122												Boron 5 **B** 10.81	Carbon 6 **C** 12.011	Nitrogen 7 **N** 14.007	Oxygen 8 **O** 15.999	Fluorine 9 **F** 18.998	Neon 10 **Ne** 20.180
Sodium 11 **Na** 22.990	Magnesium 12 **Mg** 24.305												Aluminium 13 **Al** 26.982	Silicon 14 **Si** 28.085	Phosphorus 15 **P** 30.974	Sulfur 16 **S** 32.06	Chlorine 17 **Cl** 35.45	Argon 18 **Ar** 39.948
Potassium 19 **K** 39.098	Calcium 20 **Ca** 40.078(4)		Scandium 21 **Sc** 44.956	Titanium 22 **Ti** 47.867	Vanadium 23 **V** 50.942	Chromium 24 **Cr** 51.996	Manganese 25 **Mn** 54.938	Iron 26 **Fe** 55.845(2)	Cobalt 27 **Co** 58.933	Nickel 28 **Ni** 58.693	Copper 29 **Cu** 63.546(3)	Zinc 30 **Zn** 65.38(2)	Gallium 31 **Ga** 69.723	Germanium 32 **Ge** 72.63	Arsenic 33 **As** 74.922	Selenium 34 **Se** 78.96(3)	Bromine 35 **Br** 79.904	Krypton 36 **Kr** 83.798(2)
Rubidium 37 **Rb** 85.468	Strontium 38 **Sr** 87.62		Yttrium 39 **Y** 88.906	Zirconium 40 **Zr** 91.224(2)	Niobium 41 **Nb** 92.906(2)	Molybdenum 42 **Mo** 95.96(2)	Technetium 43 **Tc** [97.91]	Ruthenium 44 **Ru** 101.07(2)	Rhodium 45 **Rh** 102.91	Palladium 46 **Pd** 106.42	Silver 47 **Ag** 107.87	Cadmium 48 **Cd** 112.41	Indium 49 **In** 114.82	Tin 50 **Sn** 118.71	Antimony 51 **Sb** 121.76	Tellurium 52 **Te** 127.60(3)	Iodine 53 **I** 126.90	Xenon 54 **Xe** 131.29
Caesium 55 **Cs** 132.91	Barium 56 **Ba** 137.33	57-70 *	Lutetium 71 **Lu** 174.97	Hafnium 72 **Hf** 178.49(2)	Tantalum 73 **Ta** 180.95	Tungsten 74 **W** 183.84	Rhenium 75 **Re** 186.21	Osmium 76 **Os** 190.23(2)	Iridium 77 **Ir** 192.22	Platinum 78 **Pt** 195.08	Gold 79 **Au** 196.97	Mercury 80 **Hg** 200.59	Thallium 81 **Tl** 204.38	Lead 82 **Pb** 207.2	Bismuth 83 **Bi** 208.98	Polonium 84 **Po** [208.98]	Astatine 85 **At** [209.99]	Radon 86 **Rn** [222.02]
Francium 87 **Fr** [223.02]	Radium 88 **Ra** [226.03]	89-102 **	Lawrencium 103 **Lr** [262.11]	Rutherfordium 104 **Rf** [265.12]	Dubnium 105 **Db** [268.13]	Seaborgium 106 **Sg** [271.13]	Bohrium 107 **Bh** [270]	Hassium 108 **Hs** [277.15]	Meitnerium 109 **Mt** [276.15]	Darmstadtium 110 **Ds** [281.16]	Roentgenium 111 **Rg** [280.16]	Copernicium 112 **Cn** [285.17]	Ununtrium 113 **Uut** [284.18]	Flerovium 114 **Fl** [289.19]	Ununpentium 115 **Uup** [288.19]	Livermorium 116 **Lv** [293]	Ununseptium 117 **Uus** [294]	Ununoctium 118 **Uuo** [294]

Key:

Element Name
Atomic number
Symbol
Atomic weight (mean relative mass)

*lanthanoids	Lanthanum 57 **La** 138.91	Cerium 58 **Ce** 140.12	Praseodymium 59 **Pr** 140.91	Neodymium 60 **Nd** 144.24	Promethium 61 **Pm** [144.91]	Samarium 62 **Sm** 150.36(2)	Europium 63 **Eu** 151.96	Gadolinium 64 **Gd** 157.25(3)	Terbium 65 **Tb** 158.93	Dysprosium 66 **Dy** 162.50	Holmium 67 **Ho** 164.93	Erbium 68 **Er** 167.26	Thulium 69 **Tm** 168.93	Ytterbium 70 **Yb** 173.05
**actinoids	Actinium 89 **Ac** [227.03]	Thorium 90 **Th** 232.04	Protactinium 91 **Pa** 231.04	Uranium 92 **U** 238.03	Neptunium 93 **Np** [237.05]	Plutonium 94 **Pu** [244.06]	Americium 95 **Am** [243.06]	Curium 96 **Cm** [247.07]	Berkelium 97 **Bk** [247.07]	Californium 98 **Cf** [251.08]	Einsteinium 99 **Es** [252.08]	Fermium 100 **Fm** [257.10]	Mendelevium 101 **Md** [258.10]	Nobelium 102 **No** [259.10]